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## **Extended DLVO Modeling of Nanoparticle-Assisted Wettability Alteration in Unconventional Reservoirs**

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### **Abstract**

Wettability alteration is a key mechanism for enhancing oil recovery in unconventional reservoirs. This study utilizes an extended Derjaguin–Landau–Verwey–Overbeek (DLVO) framework that integrates van der Waals, electrostatic double-layer (EDL), structural hydration, and nanoparticle-induced hydrophobic forces to quantify additive-assisted wettability shifts across salinity and concentration gradients. The model links total disjoining pressure to contact angle via a concentration-dependent scaling factor,  $\alpha(c)$ , and is validated against published data for tight-rock systems treated with silica nanoparticles under low-salinity water and brine conditions. Predictions match experimental contact angles with a root-mean-square error (RMSE) of  $8.97^\circ$ , capturing wettability change of rock surface in water and brine conditions under different nanoparticle concentrations. Disjoining pressure analysis indicates that EDL enhancement is the dominant mechanism of nanoparticle-induced wettability alteration, while structural hydration contributes as a short-range repulsive interaction and hydrophobic forces act as short-range attractive interactions. The framework suggests optimal nanoparticle concentrations of approximately 1 wt% in fresh water and 2 wt% in brine, providing a scalable tool for designing nanoparticle–surfactant formulations to promote spontaneous imbibition in unconventional reservoirs.

### **Introduction**

Hydrocarbons hosted in unconventional formations, including tight oil and shale gas plays, have emerged as a major component of global energy supply over the past decades (Tolmachev et al., 2020; Wang et al., 2017). These reservoirs are characterized by exceptionally low porosity (typically 2–12%) and ultra-low matrix permeability, often in the nanodarcy to microdarcy range, which severely restricts

conventional viscous-dominated flow (Jia et al., 2012). The economic viability of these resources depends on hydraulic fracturing, which creates extensive high-conductivity fracture networks that dramatically enhance connectivity between the wellbore and the formation matrix, providing pathways for hydrocarbon mobilization from previously inaccessible pore volumes. Despite the improved connectivity afforded by fractures, fluid transport within the tight matrix remains fundamentally different from conventional reservoirs (Alfarge et al., 2017). The ultra-low permeability means that pressure gradients alone cannot sustain commercial flow rates through matrix pores. Instead, capillary forces, which arise from interfacial tension and pore geometry, dominate fluid movement at the matrix scale (Yang et al., 2023). Under these conditions, spontaneous imbibition becomes the primary recovery mechanism, wherein the wetting phase (typically water or brine from fracturing fluid or injected fluids) spontaneously imbibes into the matrix pores and displaces the nonwetting hydrocarbon toward the fracture network. The efficiency of this capillary-driven process is fundamentally controlled by the rock's wettability state, which determines the balance of interfacial forces and the magnitude of the capillary driving pressure (Alvarez and Schechter, 2016).

Given this capillary-force-driven mechanism, altering reservoir rock wettability has proven to be an effective method for enhancing oil recovery (Goharzadeh and Fatt, 2022; Yuan and Dehghanpour, 2024). This wettability alteration can be chemically achieved using additives, particularly surfactants and nanoparticles, which modify interfacial properties and surface energies at the rock-fluid interface to shift oil-wet and intermediate-wet reservoirs toward more water-wet conditions. At the molecular level, these additives operate through several complementary mechanisms. First, they alter surface energies by adsorbing onto mineral surfaces and reducing oil-water interfacial tension, which facilitates the formation and stability of water films. Second, they modify electrostatic interactions at the mineral-brine interface, often strengthening electrostatic double-layer repulsion that stabilizes thin aqueous films along pore walls (Alvarez et al., 2017). Third, adsorbed surfactant layers or nanoparticle coatings introduce steric/structuring barriers and reduce surface hydrophobicity (Rezvani et al., 2024), which promotes preferential water wetting and suppresses oil re-adhesion to the rock surface. From a thin-film perspective, these modifications increase the net disjoining pressure within nanometric water films, which through the Frumkin-Derjaguin relation translates to reduced water contact angles and enhanced water-wetting behavior (Kang and Kang, 2003). The combined effect shifts the capillary pressure from opposing to favoring water imbibition, thereby enhancing spontaneous imbibition rates and improving hydrocarbon displacement efficiency from the matrix to the fracture network.

Experimental studies have demonstrated that surfactants and nanoparticles can significantly reduce contact angles and interfacial tension (IFT), leading to enhanced oil recovery through spontaneous imbibition. For example, Jiang et al. (2017) evaluated the impact of silica nanoparticles on calcite substrate wettability using nanofluids with concentrations ranging from 0.01 to 1 wt%. They found that increasing nanoparticle concentration and decreasing nanoparticle size both enhanced water-wetting behavior of the substrate (Jiang et al., 2017). A similar observation was reported by Minakov et al. (2021), who evaluated how 5–50 nm silica dioxide nanoparticles alter wettability on sandstone, dolomite, and metabasalt substrates. They found that nanoparticles increased oil contact angles (indicating a shift toward more water-wet conditions) from 73° to 153° on dolomite, from 33° to 160° on metabasalt, and from 160° to 164° on sandstone when nanoparticle concentration increased from 0 to 1 wt% (Minakov et al., 2021). Al-Anssari et al. (2018) found silica nanofluids effectively altered oil-wet calcite to a strongly water-wet state under reservoir conditions, with nanoparticle adsorption largely irreversible at high pressure and temperature (Al-Anssari et al., 2018). Wettability alteration efficiency was strongly influenced by nanofluid concentration, exposure time, and surfactant addition, while pressure and salinity showed system-dependent effects on contact angle. However, these observations are often system-specific and empirical in nature, with limited theoretical understanding of the underlying interactions. While the classical Derjaguin–Landau–Verwey–Overbeek (DLVO) theory has been widely used to model colloidal and surface interactions, it does not fully capture the complexity of additive-modified systems, particularly under high salinity or in the presence of

surface-active agents. Recent extensions to DLVO theory, incorporating additional interaction terms such as hydrophobic and steric forces, offer improved predictive capabilities but have yet to be fully integrated into practical reservoir applications.

This study addresses this gap by proposing an extended DLVO-based modeling framework that quantitatively predicts wettability alteration induced by chemical additives. The model incorporates van der Waals forces, electrostatic double-layer interactions, nanoparticle-induced steric repulsion, and hydrophobic attraction to calculate the total disjoining pressure acting across the aqueous film at the rock–oil interface. From this, the contact angle is predicted through a free energy approach, enabling a link between additive chemistry and wettability state. The model is validated using published experimental data, including contact angles and zeta potential measurements from representative unconventional systems treated with surfactants and nanoparticles. While no new experimental results are presented, the proposed framework demonstrates strong alignment with observed wettability trends across a range of salinity and additive conditions. This work contributes a scalable and physically grounded modeling tool for screening and optimizing chemical EOR formulations in unconventional reservoirs, with direct implications for additive-assisted imbibition strategies.

## Extended Disjoining Pressure Model Incorporating Nanoparticles

To predict wettability alteration in nanoparticle-assisted surfactant systems, an extended DLVO theory is employed to calculate the total disjoining pressure  $\Pi_h$ , which governs the stability of thin aqueous films at the oil–brine–rock interface. The model incorporates the classical DLVO components, van der Waals and electrostatic double-layer interactions, and augments them with nanoparticle-induced steric and hydration effects. The total disjoining pressure is expressed as (Alshakhs and Kavscek, 2016a):

$$\Pi_h = \Pi_{vdw}(h) + \Pi_{EDL}(h) + \Pi_{Structural}(h) + \Pi_{hydrophobic}(h)$$

Here  $\Pi_t$ ,  $\Pi_{vdw}$ ,  $\Pi_{EDL}$ ,  $\Pi_s$ , are the total disjoining pressure, van der Waals force, EDL force, and structural force, respectively. And  $\Pi_{hydrophobic}$  is the hydrophobic force.

**van der Waals Force.** The van der Waals force can be determined using the following equation (Israelachvili, 2011; Masliyah and Bhattacharjee, 2006):

$$\Pi_{vdw}(h) = -\frac{A_{eff}}{6\pi h^3}$$

Where  $A_{eff}$  is the effective Hamaker constant between oil and rock across the aqueous film, and  $h$  is the separation distance.

**Electrostatic Double-Layer (EDL).** The EDL component in the DLVO theory is calculated using (Israelachvili, 2011):

$$\Pi_{EDL}(h) = \frac{\epsilon \epsilon_0 \kappa^2}{2} \frac{[2\psi_1 \psi_2 \cosh(\kappa h) - \psi_1^2 - \psi_2^2]}{[\sinh(\kappa h)]^2}$$

where  $\epsilon_0$  is the vacuum permittivity ( $8.85 \times 10^{-12}$  F/m),  $\epsilon$  is the relative dielectric constant of the water (80.1 at room temperature),  $\psi_1$  and  $\psi_2$  are the surface potentials of the two interfaces,  $\kappa$  is the inverse Debye length, and  $\kappa^{-1}$  is Debye length.

**Structural Force.** The structural force due to the hydration layer can be determined by (Derjaguin and Zorin, 1992):

$$\Pi_{Structural}(h) = A_s e^{-\frac{h}{h_s}}$$

Where  $A_s$  and  $h_s$  are widely assumed to be  $1.5 \times 10^{10}$  Pa, and 0.05 nm respectively (Alshakhs and Kavscek, 2016b; Sanaei et al., 2019).

**Hydrophobic force.** In the presence of surface-modified nanoparticles with partial hydrophobic character, additional short-range hydrophobic forces can arise from the entropy-driven rearrangement of water molecules near nanoparticle-covered surfaces. This effect decays exponentially with the separation distance. If only short-range interactions are considered, the hydrophobic component can be expressed as (Israelachvili and Pashley, 1982):

$$\Pi_{hydrophobic}(h) = -Ce^{-\frac{h}{\lambda}}$$

where  $C$  is the structural force coefficient and  $\lambda$  is the characteristic decay length. Previous study (Churaev and Sobolev, 2007; Yang et al., 2015) reported that  $C$  typically ranges from  $10^5$  to  $10^7$  Pa, with  $\lambda$  between 0.1–0.3 nm. In this study, both  $C$  and  $\lambda$  are determined through data fitting to match the measured wettability trends.

### Free Energy and Contact Angle Calculation

The total disjoining pressure is integrated to obtain excess interfacial free energy (Basu and Sharma, 1996; Sanaei et al., 2019).

$$\Delta G = \int_{h_0}^{\infty} \Pi_h dh$$

This excess free energy is then related to the contact angle  $\theta$  via the modified Young–Dupré equation (Basu and Sharma, 1996; Churaev and Sobolev, 1995):

$$\cos \theta = 1 + \alpha \frac{\Delta G}{\gamma_{ow}}$$

Here,  $\alpha$  is considered a function of nanoparticle concentration ( $\alpha = \alpha_0 + \alpha_1 \times C_{NP}$ ), allowing the model to more accurately reproduce the experimental data for TW+NP and brine+NP systems at different nanoparticle concentrations. The parameters  $\alpha_0$  and  $\alpha_1$  are obtained by fitting the model to experimental measurements.

## Experimental Data

To validate the proposed extended DLVO-based wettability model, we utilize experimental data previously published by Yuan et al (2021) that systematically examined nanoparticle-assisted wettability alteration in tight rock systems. The experimental program includes contact-angle measurements, zeta potential tests, and interfacial tension evaluations conducted on siltstone core samples from the Montney Formation, a well-characterized unconventional reservoir in Western Canada. The oil and fluid properties and components were also reported in their study.

### Experimental System and Additives

Rock samples were extracted from the lower Montney Formation in Canada and exhibited an average porosity of 4.51% and gas permeability of 9.02  $\mu$ D. The aqueous phase consisted of either tap water (TW) or formation brine, both of which were used to formulate nanoparticle suspensions with surface-modified silica nanoparticles at concentrations of 1 wt%, 2 wt%, and 7 wt%. Condensate oil from the same formation was used as the non-aqueous phase in all tests.

### Contact Angle Measurements

Static liquid–liquid contact angles (CA) were measured using the sessile droplet method under two experimental modes. In the equilibrium tests, rock samples were pre-soaked in low-salinity tap water (TW), brine, or NP solutions and then exposed to oil droplets. In the dynamic tests, NP solution was introduced into a brine-saturated environment to observe CA changes driven by NP diffusion. The baseline TW and brine systems yielded oil-wet conditions, with CA values of approximately 94°–97°, whereas NP-treated

fluids significantly reduced the CA to below 60°, indicating a shift toward more water-wet states. In TW, a nanoparticle concentration of 1 wt% was sufficient to achieve maximal wettability alteration, with higher concentrations offering diminishing returns.

### Zeta Potential and Interfacial Tension Data

Zeta potential ( $\zeta$ ) measurements were performed for both rock–water and oil–water systems. The presence of nanoparticles increased the absolute value of  $\zeta$ , reflecting more stable electrostatic interactions, particularly under low-salinity conditions. IFT between oil and aqueous phases decreased with increasing NP concentration, with a minimum of 0.190 mN/m observed at 7 wt% NP in brine.

### Dataset Utilization for Model Validation

In this study, the extended DLVO model is parameterized and validated using previously published measurements, as shown in Table 1. Zeta potential data are used to define surface potentials in the EDL term, while IFT values are incorporated into the Young–Dupré equation to relate disjoining pressure to contact angle. Measured contact angles serve as benchmarks for comparing model predictions under different salinity and nanoparticle concentration conditions. No new experiments were conducted; all data for model calibration and validation are derived from Yuan et al. (2021) and are representative of field conditions in unconventional tight oil systems. The model's ability to reproduce experimental trends confirms its applicability for screening and optimizing additive-assisted wettability modification strategies. For all calculations, the effective Hamaker constant between oil and rock across the aqueous film is set to  $2.49 \times 10^{21} \text{ J}$ .

**Table 1—Summary of experimental conditions and key measurements used for model parameterization and validation, including nanoparticle concentration ( $C_{NP}$ ), salinity, Debye length ( $\kappa^{-1}$ ), surface potentials ( $\psi_1$ ,  $\psi_2$ ), measured contact angle, zeta potential, and interfacial tension (IFT). Wettability interpretation is based on the contact angle classification, with lower CA values indicating more water-wet conditions.**

| Test Type   | Condition         | $C_{NP}$<br>(wt%) | Salinity   | Debye length<br>( $\kappa^{-1}$ ) | $\psi_1$ | $\psi_2$ | Measured<br>CA (°) | IFT<br>(mN/m) | Wettability<br>Interpretation |
|-------------|-------------------|-------------------|------------|-----------------------------------|----------|----------|--------------------|---------------|-------------------------------|
| Equilibrium | TW (base case)    | 0                 | 0 ppm      | 2.39E-09                          | -18      | -50      | ~97                | 2.25          | Strong oil-wet                |
|             | Brine (base case) | 0                 | 13,000 ppm | 1.67E-09                          | -5       | -21      | ~94                | 2.49          | Strong oil-wet                |
|             | TW + NP           | 1                 | 0 ppm      | 2.39E-09                          | -38      | -60      | ~52                | 0.447         | Water-wet achieved            |
|             | TW + NP           | 2                 | 0 ppm      | 2.39E-09                          | -37      | -65      | ~54                | 0.385         | Slightly less water-wet       |
|             | TW + NP           | 7                 | 0 ppm      | 2.39E-09                          | -37      | -63      | ~58                | 0.344         | Slightly less water-wet       |
|             | Brine + NP        | 1                 | 13,000 ppm | 1.67E-09                          | -16      | -38      | ~67                | 0.267         | Moderate wettability shift    |
|             | Brine + NP        | 2                 | 13,000 ppm | 1.67E-09                          | -17      | -40      | ~62                | 0.230         | Improved wettability          |
|             | Brine + NP        | 7                 | 13,000 ppm | 1.67E-09                          | -15.5    | -41      | ~65                | 0.190         | Slight recovery of oil-wet    |
| Dynamic     | Brine → +NP       | 0 → 0.5           | 13,000 ppm | 1.67E-09                          | -        | -        | 95 → 57 (after 6h) | ↓ over time   | Gradual wettability shift     |



## Results and Discussions

This section summarizes the model performance, parameter interpretation, and wettability trends under varying salinity and nanoparticle concentrations, linking predicted disjoining pressure behavior to measured contact angle shifts.

### Model Performance and Validation

Figure 1 compares the predicted and measured contact angles for the optimal parameter set obtained from grid search. Using the dataset in Table 1, the model achieved a root-mean-square error (RMSE) of  $9.34^\circ$ , an  $R^2$  of 0.7, and a mean absolute error (MAE) of  $7.8^\circ$ , demonstrating strong agreement with experiments. The validation covers contact angles from moderately oil-wet ( $\sim 97^\circ$ ) to water-wet ( $\sim 52^\circ$ ), with nanoparticle concentrations ranging from 0 to 7 wt% in both deionized water (TW) and high-salinity brine (13,000 ppm NaCl). Most predictions fall close to the 1:1 reference line, with larger deviations at higher contact angles ( $>80^\circ$ ) likely due to uncertainties in oil-wet CA measurements or model simplifications such as constant Hamaker constants.

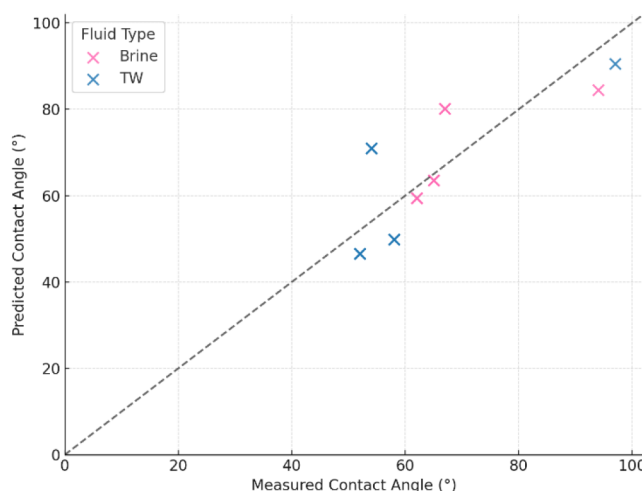


Figure 1—Comparison between predicted and measured CA for TW and brine (13,000 ppm) systems with nanoparticle concentrations ranging from 0 to 7 wt%. The dashed line represents the 1:1 agreement.

### Interpretation of Optimal Parameters

The optimal parameter set obtained from grid search fitting of the extended DLVO model to the measured contact angle data is summarized in Table 2. The characteristic separation distance  $h_0$  was slightly larger for brine (0.4–0.5 nm) than for TW (0.3–0.4 nm). The fitted hydrophobic coefficient  $C$  of  $1.0 \times 10^7$  Pa lies toward the upper end of reported literature values, indicating a pronounced short-range structural interaction between the oil and rock surfaces. The decay length  $\lambda$  of 0.3 nm reflects the localized nature of this force.

Table 2—Optimal parameter set obtained from grid search fitting of the extended DLVO model to measure contact angle data.

| Parameter                          | Symbol         | TW Value          | Brine Value   |
|------------------------------------|----------------|-------------------|---------------|
| Hydrophobic coefficient            | $C$ (Pa)       | $1.0 \times 10^7$ | same          |
| Decay length (hydrophobic)         | $\lambda$ (nm) | 0.3               | same          |
| Characteristic separation distance | $h_0$ (nm)     | 0.4               | 0.5           |
| $\alpha(c)$ baseline               | $a_0$          | 1.989             | 2.270         |
| $\alpha(c)$ slope                  | $a_1$          | −5.23             | −12.80        |
| RMSE ( $^\circ$ )                  | —              | 8.97 $^\circ$     | 8.97 $^\circ$ |

The  $\alpha(c)$  baseline values ( $\alpha_0$ ) were 1.98 for TW and 2.27 for brine, both exceeding 1. This suggests that in the absence of nanoparticles, theoretical excess free energy  $\Delta G$  must be scaled upward by roughly a factor of two to match the experimental contact angles. This scaling may compensate for underestimation of EDL forces in the theoretical formulation or additional contributions from mineralogical heterogeneity and surface roughness not captured by the idealized DLVO framework. Negative slope parameters ( $a_1$ ) for both systems confirm that nanoparticle addition enhances the repulsive contributions to the total disjoining pressure, thereby lowering the contact angle. The steeper slope in brine (-12.80) compared to TW (-5.23) indicates greater sensitivity to nanoparticle concentration in high-salinity environments, although this enhancement may diminish rapidly with increasing dosage due to salinity-induced screening effects.

### Interpretation of $\alpha(c)$ Trends

Figure 2 shows the variation of the scaling parameter  $\alpha(c)$  with nanoparticle concentration for TW and brine systems. In both cases,  $\alpha$  decreases approximately linearly as  $C_{NP}$  increases, indicating that higher nanoparticle loadings reduce the effective free energy contribution to wettability alteration relative to the theoretical  $\Delta G$  predicted by the extended DLVO model.

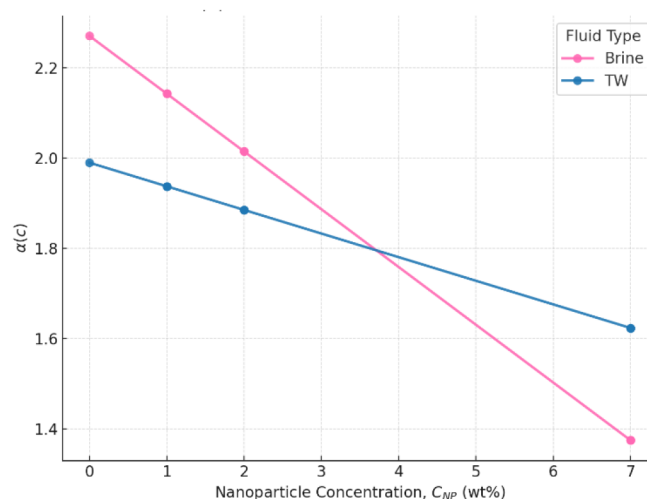


Figure 2—Measured and predicted contact angles as a function of nanoparticle concentration for TW (blue) and brine (pink) systems. Solid lines represent model predictions, and open markers represent experimental measurements.

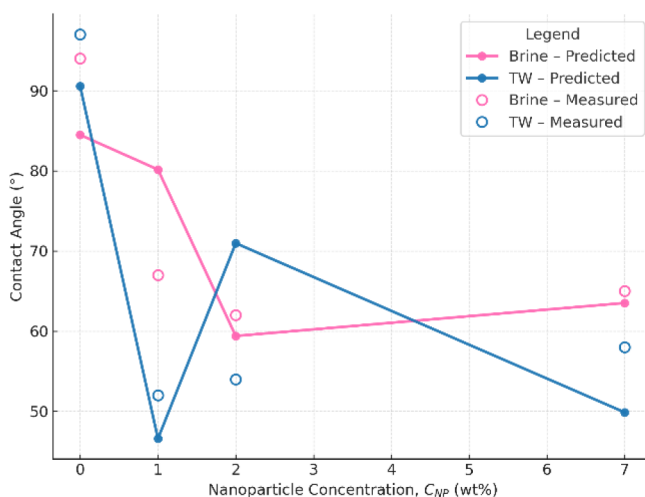
At zero NP concentration ( $C_{NP}=0$ ),  $\alpha_0$  values of 1.99 for TW and 2.27 for brine highlight the need to amplify the theoretical prediction to match experimental results. The faster decline of  $\alpha(c)$  in brine is attributed to multiple factors: (i) the shorter Debye length at high ionic strength reduces the EDL repulsion range; (ii) nanoparticles are more prone to aggregation in saline conditions, decreasing their effective surface coverage; and (iii) cation bridging and enhanced charge screening in brine further reduce NP-induced hydrophilicity.

By  $C_{NP}=7$  wt%,  $\alpha(c)$  drops to 1.62 for TW and 1.38 for brine, confirming the sharper performance decline in high-salinity environments. Practically, this suggests that low-salinity systems can tolerate higher nanoparticle dosages without a substantial loss in wettability alteration efficiency, whereas high-salinity systems require careful optimization to avoid diminishing returns.

### Effect of Salinity and NP Concentration on Wettability

The combined influence of salinity and nanoparticle concentration on wettability alteration is illustrated in Figure 3. In TW systems, increasing  $C_{NP}$  from 0 to 1 wt% leads to a significant reduction in contact angle, indicating a clear shift toward more water-wet conditions. Beyond approximately 2 wt%, the rate of improvement diminishes, but no adverse wettability reversal is observed up to the tested maximum of 7 wt

%. In contrast, brine systems exhibit a much steeper initial reduction in contact angles at low NP dosages, followed by a plateau and, in some cases, a slight reversal at higher concentrations (>4 wt%). This behavior is consistent with the strong negative slope of  $\alpha(c)$  in brine (-12.80), which accelerates the decline in the effective driving force for wettability alteration. The sharper decline in brine may be attributed to several factors. First, the shorter Debye length at high ionic strength reduces EDL repulsion. Second, nanoparticles tend to aggregate more readily in saline environments, which decreases the effective surface coverage on the rock surface. Third, stronger cation bridging and surface charge screening in brine further diminish the NP-induced wettability change.



**Figure 3—Measured and predicted contact angles as a function of nanoparticle concentration for TW (blue) and brine (pink) systems. Solid lines represent model predictions, and open markers represent experimental measurements. The model captures the overall decreasing trend of contact angles with increasing nanoparticle concentration in both low- and high-salinity systems, with diminishing returns at higher concentrations—particularly in brine.**

Quantitatively, the model predicts that at 1 wt% NP concentration, the contact angle reduction is approximately 12–15° in TW and 18–20° in brine relative to NP-free conditions. However, at 7 wt% NP, the net reduction compared to the baseline is ~20° in TW but only ~13° in brine, demonstrating the diminishing returns of high NP loadings in saline environments. These observations imply that low-salinity injection fluids can accommodate moderate to high NP concentrations without significant loss of performance, whereas high-salinity systems require a carefully optimized NP dosage—likely in the range of 0.5–2 wt%—to avoid performance decline.

### Force Contribution Analysis

The decomposition of disjoining pressure into van der Waals, electrostatic double layer, hydrophobic, and structural components provides an interpretation of the observed wettability changes. Figure 5 summarizes the force profiles for TW and brine systems under baseline conditions (NP-free) and at three representative nanoparticle concentrations of 1 wt%, 2 wt%, and 7 wt%. In the NP-free baseline, the total  $\Pi(h)$  is strongly negative across the 1–10 nm separation distance range, indicating net attraction between the rock and oil phases. This attraction is dominated by vdW forces, with only partial compensation from EDL repulsion. The magnitude of EDL is notably smaller in brine than in TW, reflecting the shorter Debye length and stronger charge screening in high-salinity environments. Consequently, the brine system starts from a more oil-wet baseline state.

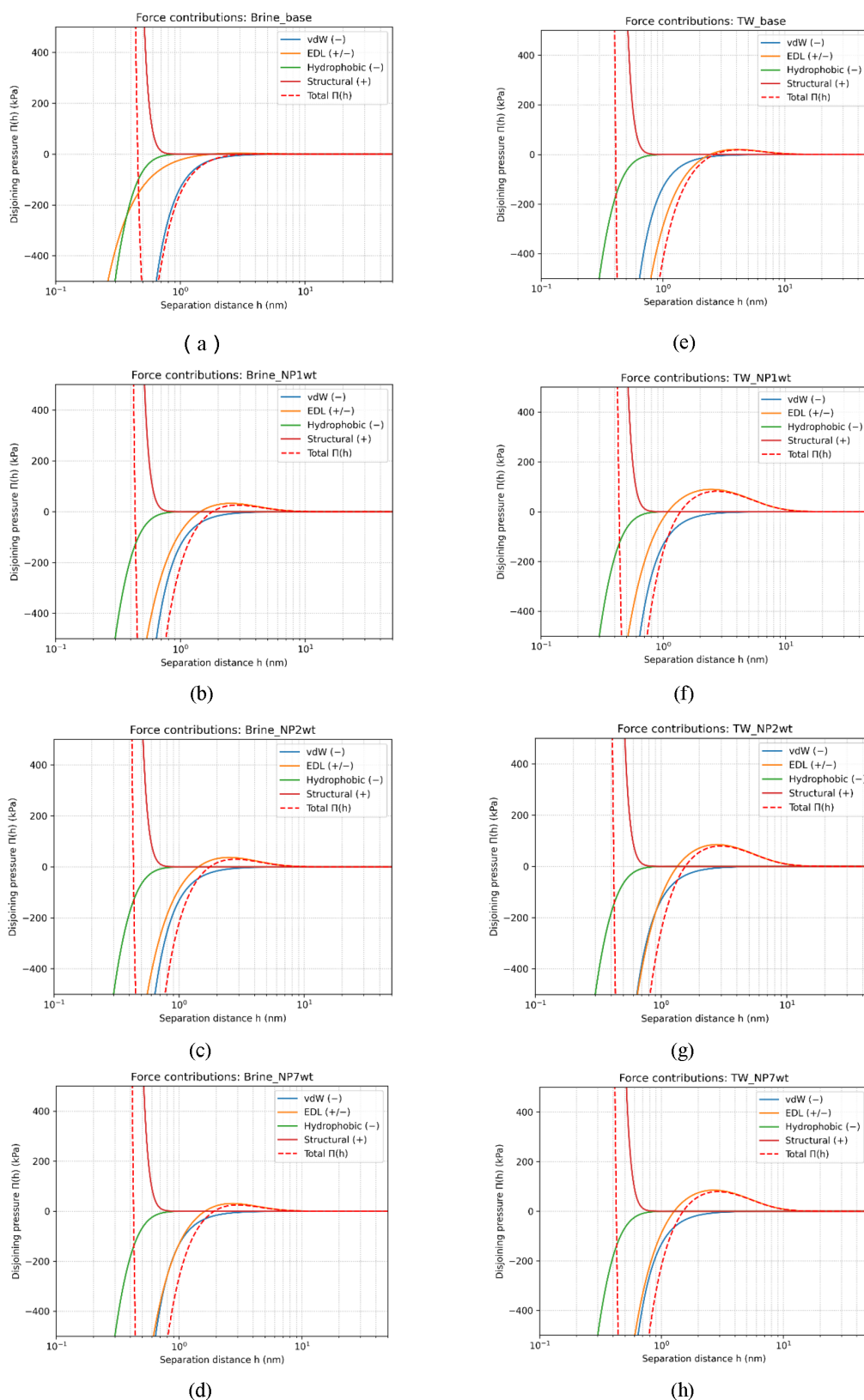
At very short separation distances ( $h < 0.3$  nm),  $\Pi(h)$  is strongly positive for all cases due to short-range structural (hydration) repulsion between the rock and aqueous phase. This positive contribution decays exponentially with increasing separation distance and becomes negligible beyond ~0.3 nm, after which



van der Waals attraction dominates unless countered by EDL repulsion. On the other hand, hydrophobic interactions remain short-ranged and attractive in all cases (Dehghan Monfared et al., 2018).

Upon the addition of nanoparticles, the force balance changes primarily through EDL enhancement, which increases short-range repulsion and shifts  $\Pi(h)$  toward less negative values. Both TW and brine systems exhibit a pronounced repulsive peak slightly beyond 1 nm separation. The magnitude of this peak is consistently higher in TW than in brine, reflecting more extended EDL interactions in low-salinity conditions. Furthermore, NP-treated systems show higher peak values than their NP-free counterparts, indicating enhanced repulsion due to nanoparticle adsorption.

At optimal NP concentrations (1–2 wt%), this EDL-driven repulsion is sustained over a wider separation range in TW, leading to greater positive shifts in the integrated excess free energy change ( $\Delta G$ ) and, via the relationship  $\cos \theta = 1 + \alpha \frac{\Delta G}{\gamma_{ow}}$ , lower contact angles and more water-wet conditions. In brine, although the initial EDL peak increases with NP addition, its magnitude decays more rapidly with distance, limiting its contribution to  $\Delta G$ . At higher NP loadings (7 wt%), the benefit plateaus in both systems, and in brine may be further reduced due to particle aggregation and surface coverage loss. These observations confirm that nanoparticle-induced wettability alteration is governed primarily by EDL enhancement, with salinity exerting a strong control on both the magnitude and persistence of this effect.



**Figure 4—Decomposition of disjoining pressure  $\Pi(h)$  into van der Waals (blue), EDL (orange), hydrophobic (green), structural (red dashed), and total (red solid) components for brine (left column) and TW (right column) systems. (a,e) NP-free baseline; (b,f) 1 wt% NP; (c,g) 2 wt% NP; (d,h) 7 wt% NP. Positive values denote repulsive contributions. NP addition enhances short-range EDL and structural repulsion, shifting total  $\Pi(h)$  toward less negative values and promoting more water-wet conditions, especially in TW systems.**

## Conclusion

This study validated an extended DLVO framework, incorporating hydrophobic and structural hydration contributions, to quantify nanoparticle-assisted wettability alteration across salinity gradients. The model accurately reproduced measured contact angles for both tap water and brine systems, with a root-mean-square error (RMSE) of  $8.97^\circ$  and provided physically interpretable parameters that capture the interplay between van der Waals, electrostatic double-layer (EDL), and short-range hydration and hydrophobic forces.

Disjoining pressure analysis revealed that EDL enhancement is the primary driver of wettability alteration in the presence of nanoparticles, while both structural hydration and hydrophobic forces remain short-ranged. At separations below  $\sim 0.3$  nm, structural hydration yields a strong positive repulsion; beyond this range, EDL dominates the repulsive contribution, and its magnitude is notably higher in TW than in brine due to the longer Debye length and weaker charge screening. Nanoparticle addition increases the EDL peak and shifts the total disjoining pressure toward less negative values, with the enhancement being more pronounced in TW than in brine. The  $\alpha(c)$  scaling approach effectively captured the concentration-dependent decline in wettability alteration efficiency, with TW showing gradual reduction in  $\alpha$  and brine exhibiting a steeper decline. From an application perspective, low-salinity systems benefit from moderate NP concentrations ( $\sim 1$  wt%), achieving substantial wettability alteration with minimal chemical usage. High-salinity systems require more careful dosage control, as concentrations exceeding 2 wt% offer limited additional benefits and may incur unnecessary costs. The proposed methodology advances the understanding of additive-assisted wettability modification and offers a transferable tool for optimizing nanoparticle–surfactant treatments under different salinity conditions in unconventional reservoirs.

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